$$2B + 3Cl_2 \longrightarrow 2BCl_3$$

$$2S + Cl_2 \longrightarrow \underset{\text{disulphur dichloride}}{S_2Cl_2}$$

$$P_4 + 6Cl_2 \longrightarrow 4PCl_3$$

$$2As + 3Cl_2 \longrightarrow 2AsCl_3$$

$$2Sb + 3Cl_2 \longrightarrow 2SbCl_3$$

Affinity for hydrogen : When burnt with turpentine it forms carbon and hydrochloric acid.

$$C_{10}H_{16} + 8Cl_2 \longrightarrow 10C + 16HCl$$

It forms dioxygen when reacting with water in presence of sunlight. When chlorine in water is exposed to sunlight it loses its colour and smell as the chlorine is converted into hydrochloric acid.

$$2Cl_2 + 2H_2O \longrightarrow O_2 + 4HCl$$

Chlorine reacts with ammonia to give ammonium chloride and other products as shown below:

With excess ammonia,

$$2NH_3 + 3Cl_2 \longrightarrow N_2 + 6HCl$$

$$6HCl + 6NH_3 \longrightarrow 6NH_4Cl$$

overall reaction

$$8NH_3 + 3Cl_2 \longrightarrow N_2 + 6NH_4Cl$$

With excess chlorine,

$$NH_3 + 3Cl_2 \longrightarrow NCl_3 + 3HCl$$

$$3HCl + 3NH_3 \longrightarrow 3NH_4Cl$$

overall reaction

$$4NH_3 + 3Cl_2 \longrightarrow NCl_3 + 3NH_4Cl$$

Chlorine oxidises hydrogen sulphide to sulphur and liberates bromine and iodine from iodides and bromides. However, it doesn't oxidise fluorides

$$H_2S + Cl_2 \longrightarrow 2HCl + S$$

$$Cl_2 + 2KBr \longrightarrow 2KCl + Br_2$$

$$Cl_2 + 2KI \longrightarrow 2KCl + I_2$$

Reaction with alkali: Chlorine reacts with cold dilute alkali to give chloride and hypochlorite while with hot concentrated alkali chlorides and chlorates are formed.

$$Cl_2 + H_2O \longrightarrow HCl + HOCl$$

$$HCl + NaOH \longrightarrow NaCl + H_2O$$

$$HOCl + NaOH \longrightarrow NaOCl + H_2O$$

overall reaction

$$Cl_2 + 2NaOH \longrightarrow \underset{\text{sodium hypo chlorite}}{NaOCl} + NaCl + H_2O$$

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